

Ethan Rosenfeld

ethan.rosenfeld@duke.edu | (917) 885-5807

[linkedin.com/in/ethan-rosenfeld06](https://www.linkedin.com/in/ethan-rosenfeld06) | github.com/contraflux

Education

Duke University, Durham, NC

Expected May 2029

GPA: 4.0

Majors: Mechanical Engineering, Physics

Certificates: Aerospace Engineering

Horace Mann High School, Bronx, NY

June 2025

Henry J. Binder, M.D. '53 Scientific Achievement Award in Physics (12)

Robotics, Computer Science, and Engineering Student of Distinction Award (12)

Work Experience

Mid Island Air Service, *Aircraft Maintenance (Intern)*, Shirley, NY

Summer 2022, Summer 2023

- Performed annual and 100-hour inspections alongside airframe and powerplant mechanics on Beechcraft, Cessna, and Piper aircraft.
- Assisted in engine maintenance, including cylinder compression tests, servicing spark plugs, changing oil filters and fluid lines, and replacing and timing magnetos
- Serviced flight control system, seats, and interior carpeting

Extracurricular Activities

Duke Aero, *Propulsion Engineer*, Durham, NC

2025-Present

- Building Duke Aero's first liquid rocket motor
- Researched and mixed propellants for solid rocket motor
- Designed and optimized the team's solid rocket motor nozzle

MIT BWSI Unmanned Aerial System Synthetic Aperture Radar, Cambridge, MA

Summer 2024

- Wrote Python scripts for a Raspberry Pi to collect data from a PulsON 440 radar
- Designed and interpreted Range-Time-Intensity plots from collected data
- Created Synthetic Aperture Radar maps by writing a backprojection algorithm in Python
- Successfully detected landmines buried in sand using Synthetic Aperture Radar in collaboration with MIT Lincoln Laboratory

FRC Robotics Teams 5806, 10452, *Electronics Team Lead*, Bronx, NY

2023-2025

- Wrote a Java program to detect and align the robot to QR codes
- Wrote Java code to coordinate multiple actions to facilitate robot scoring for the operator
- Designed and organized electrical component layout

Aviation Challenge Mach III, Huntsville, AL

Summer 2022

- Flag Award winner, awarded to a team for exemplary teamwork and attention to detail
- Top Gun winner, awarded to an individual for winning the simulated dogfight tournament

Technical Projects

Math and Physics Demonstrations (JavaScript, HTML, CSS)

2024-Present

- Web apps to demonstrate concepts in math and physics
- Projects include vector fields, circuits, electric fields, and matrix transformations.

Electromagnetism (Julia)

2024-Present

- Julia script to calculate and display electric and magnetic fields given a configuration of moving charges

Flight Simulator Asset Design (Photoshop)

2023-Present

- Created various airport sceneries and designed liveries for aircraft.

Technical Skills

Certificates: Private Pilot License, Instrument Rating

Languages: Python, Java, JavaScript, Julia, HTML, CSS

Software: MATLAB, Ansys Fluent, CAD, Git, Photoshop

Hardware: Raspberry Pi, Arduino, Built a computer